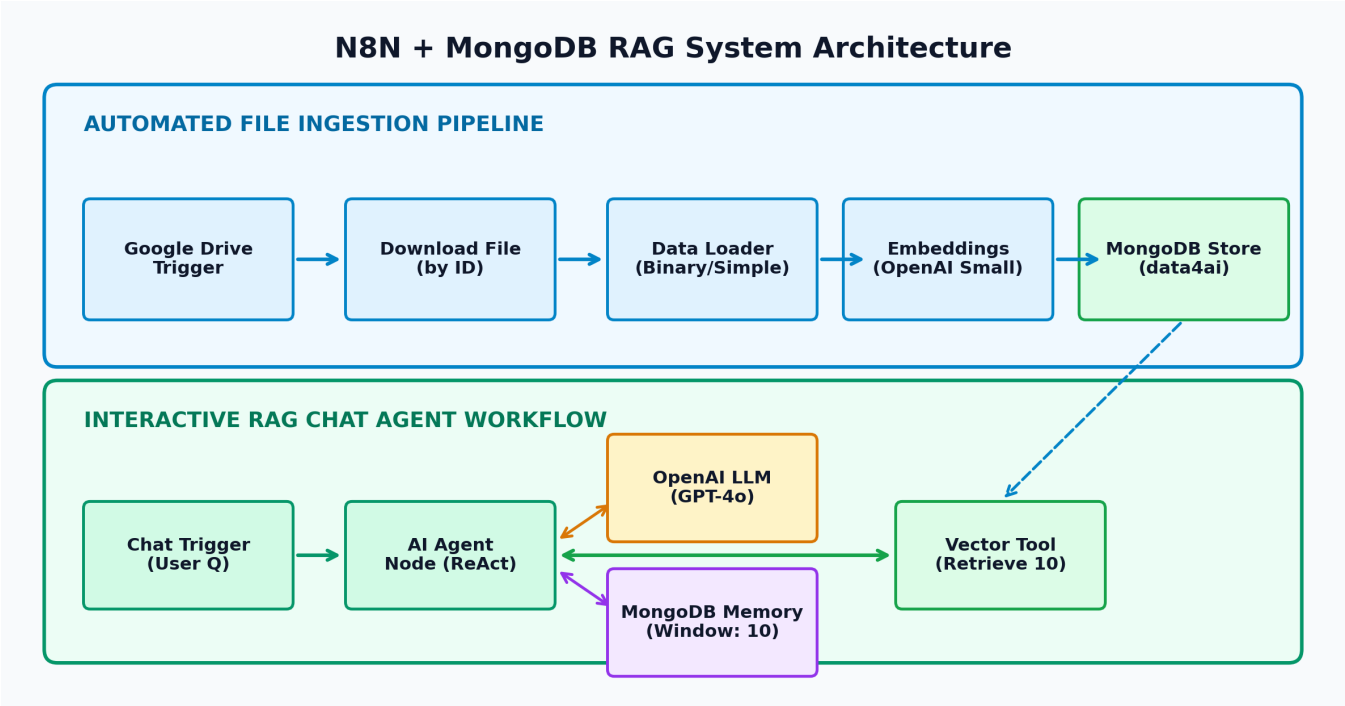
How to Build a RAG-Powered AI Agent with n8n + MongoDB

Complete Production Architecture, Workflow Configurations, Prompts & Testing Guide

1. Executive Summary & Tech Stack Overview

Retrieval-Augmented Generation (RAG) is essential for AI agents requiring access to large external datasets, spreadsheets, and document archives. Connecting LLM agents to a production vector store eliminates context window limits and guarantees accurate responses. This guide breaks down the complete architecture to deploy a no-code, production-ready RAG agent using **n8n** for workflow orchestration and **MongoDB Atlas** for vector storage and chat memory.

Component	Technology / Tool	Role & Configuration in Workflow
Orchestrator	n8n (No-Code)	Workflow triggers, node-based routing, and tool integration.
Vector Store	MongoDB Atlas	Cloud database hosting vector search index (rag_vector_index).
Chat LLM	OpenAI GPT-4o	Reasoning engine executing ReAct loop and answering queries.
Embeddings	text-embedding-3-small	Converts text into 1536-dimensional vector representations.
Chat Memory	MongoDB Chat Memory	Persists state across turns (context window length = 10 messages).
Ingestion Trigger	Google Drive Node	Monitors folder for new CSV or PDF file uploads automatically.



2. Step-by-Step MongoDB Atlas Vector Store & Index Setup

MongoDB Atlas serves as both the document vector database and persistent chat memory store. The following steps outline cloud cluster deployment and vector index initialization:

• **1. Create Deployment:** Deploy a free cluster named VectorStore on AWS in your closest region.

• **2. User & Network Security:** Create a database user with password auth. Under Network Access, whitelist IP access (e.g. 0.0.0.0/0 for n8n cloud).

• **3. Connection String:** Copy driver connection string (mongodb+srv://<username>:<password>@vectorstore...).

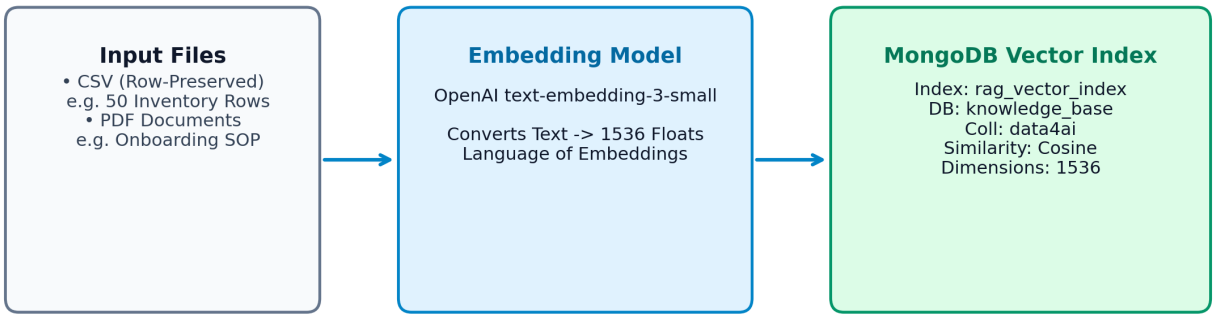
• **4. Create Database & Collection:** In Browse Collections, create Database: knowledge_base and Collection: data4ai.

• **5. Create Search Index:** Navigate to Atlas Search -> Create Search Index -> **Vector Search**. Configure index name: rag_vector_index.

CRITICAL MONGO VECTOR INDEX CONFIGURATION PARAMETERS

• **Index Name:** rag_vector_index
• **Target Database & Collection:** knowledge_base.data4ai
• **Vector Field Path:** embedding
• **Dimensions:** 1536 (Exact match for OpenAI text-embedding-3-small model)
• **Similarity Metric:** cosine (Optimized for semantic similarity lookup)

MongoDB Vector Index & Embeddings Processing Flow



Key Advantage: Row integrity in CSVs ensures full metadata (Price, Qty, Supplier) is kept in a single chunk.

3. Interactive RAG Chat Agent Node Configuration & Prompts

The primary workflow handles incoming chat messages, loads memory context, queries MongoDB Atlas Vector Store when needed, and produces grounded answers.

Node Name	Node Type	Key Parameters & Instructions
Chat Trigger	On Chat Message	Triggers workflow upon receiving user query via n8n chat UI.

AI Agent	Tools Agent (ReAct)	Connects LLM, Memory, and Vector Tool. Configured with System Prompt.
Language Model	OpenAI Chat Model	Model: gpt-4o (handles tool calling and reasoning).
Chat Memory	MongoDB Chat Memory	DB: knowledge_base, Collection: chat_histories, Context Window: **10** messages.
Knowledge Tool	MongoDB Vector Store	Operation: *Retrieve Documents as Tool*. Index: rag_vector_index, Retrieval Limit: **10**.
Embeddings	Embeddings OpenAI	Model: text-embedding-3-small (1536 dims).

Recommended System Message & Tool Prompts

PRODUCTION PROMPT TEMPLATES FOR LEARNING

System Message (AI Agent Node):
"You are an intelligent business assistant. Answer the user's questions accurately using the knowledge_base tool whenever you require external information or document data. If you do not know the answer, explicitly invoke the tool before responding."

Tool Description (MongoDB Atlas Vector Store Node):
"Call this tool to access your knowledge base containing inventory CSVs, client SOPs, and internal documents."

4. Google Drive Ingestion Pipeline (Auto Ingest CSV & PDF)

To allow non-technical users to update the agent's knowledge base without coding, a second automated sub-workflow monitors a designated Google Drive folder.

• **1. Google Drive Trigger Node:** Event: *Changes involving a specific folder*. Set file event to *File Created* in folder Knowledge Base RAG.

• **2. Download File Node:** Download file binary content by referencing file ID dynamically from trigger node ({{ $json.id }}).

• **3. MongoDB Vector Store (Add Docs):** Operation: *Add Documents to Vector Store*. Connects to Embeddings OpenAI and a Default Data Loader.

• **4. Data Loader Configuration:** Type: Binary, Data Field: data, Mode: Load specific data. Text splitting: Simple.

5. Verification, Real-World Testing & Scenarios

The system was validated using two distinct document types: an inventory tracking CSV (50 records) and a Client Onboarding SOP PDF.

Test Type	Input File / Source	Test User Prompt	Agent Outcome & Verification
CSV Row Integrity	Inventory CSV (50 records)	*"List all items supplied by Netgear"*	Accurately retrieved all 6 items (IDs 1, 17, 19, 31, 34, 43). Preserved row metadata.

Exact Detail Query	Inventory CSV	*"Give me all details for IDM005"*	Extracted exact item specs: Printer, Qty: 96, Price: $21.56.
Memory Retention	MongoDB Memory	*"What was my last question?"*	Successfully recalled prior Netgear query from MongoDB 10-message window.
PDF Chunking	Client Onboarding SOP (PDF)	*"What is the step-by-step process for onboarding?"*	Parsed 5 PDF vector chunks to output Welcome Email, Internal Handoff, SOP steps.
Tool Extraction	Client Onboarding SOP (PDF)	*"What tools are required?"*	Correctly listed HubSpot, Gmail, Google Drive, ClickUp, Slack, Zoom.

6. Key Architectural Takeaways for Production Deployment

1. Row Integrity in CSV Ingestion: Unlike naive text splitters that break rows across chunks, MongoDB Atlas vector ingestion maintains single CSV rows in unified documents, preserving field relationships (e.g. price linked to item ID).

2. State Retention via MongoDB Memory: Storing chat histories in MongoDB Atlas rather than volatile in-memory storage provides persistent session state and allows auditing user conversations.

3. Retrieval Limit Optimization: Increasing vector retrieval limit (e.g. from 4 to 10) allows the LLM to analyze broader datasets without missing target records.

4. Separation of Concerns: Google Drive auto-ingestion runs independently from chat execution, keeping the chat interface responsive.